PUSL3190 Computing Project

Project Initiation Document (PID)

Pomegranate Ripeness and Disease Detection System

Supervisor: Mr. Gayan Perera

Name: SDK Dananjaya
Plymouth Index Number: 10953479
Degree Program: BSc Computer Science

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Chapter 01- Introduction

1.1. Purpose and Scope

The Pomegranate Ripeness and Disease Detection System project addresses one of critical challenge in Sri Lankan agriculture which is the need for accessible, offline fruit quality assessment technology for small scale farmers. This project aims to develop a lightweight, real-time mobile application integrated with optimized machine learning models to enable farmers to determine fruit ripeness and detect diseases without relying on expensive infrastructure or constant internet connectivity.

1.2. Background and Context

Agriculture forms the backbone of Sri Lanka's economy with fruit cultivation contributing significantly to both local food production and export value. Pomegranate (Punica granatum) has emerged as a high value fruit due to its nutritional benefits and growing market demand. However, the lack of accurate ripeness assessment and early disease detection tools forces farmers to rely on traditional manual inspection methods which are time consuming, labor intensive and prone to human error.
Sri Lankan pomegranate varieties such as Lanka Red and Malee Pink have unique characteristics distinct from globally studied cultivars. Current agricultural automation solutions depend heavily on cloud based processing or high-end hardware making them unaffordable and impractical for rural farming communities where internet connectivity is limited and computational resources are constrained.

1.3. Problem Statement

Fruit quality assessment in Sri Lanka currently relies on manual inspection methods that are inefficient and error prone. Farmers, particularly those in rural regions with limited access to advanced technologies experience significant financial losses due to improper harvesting timing, poor storage decisions and undetected diseases that spread rapidly within crops. Additionally, there is a critical absence of localized datasets and automated tools specifically adapted to Sri Lankan pomegranate varieties further reducing the accuracy and applicability of existing machine learning models.

1.4. Scope and Limitations

Included:

- On-device machine learning models for pomegranate fruit detection, ripeness classification and disease identification
- Custom dataset addition for development featuring Sri Lankan pomegranate varieties (Lanka Red, Malee Pink etc.)
- Dataset augmentation and preprocessing techniques
- Flutter mobile application with real-time camera-based inference
- TensorFlow Lite optimization and performance benchmarking

- Augmented reality (AR) visualization overlays on live camera feed

Excluded:

- Cloud-based inference or server-side processing
- Disease detection beyond visually observable symptoms
- Enterprise level production deployment

1.5. Expected Impact and Stakeholders

Primary Stakeholders:

- Small scale farmers in rural Sri Lanka seeking efficient fruit quality assessment solutions
- Department of Agriculture officials collaborating on dataset development and validation
- Agricultural cooperatives looking to modernize farming practices

Expected Impact:

- Reduction in harvest losses through improved ripeness assessment accuracy
- Improved fruit quality control and market competitiveness
- Enhanced farmer productivity through faster decision making
- Demonstration of practical, low-cost AI application in agricultural contexts
- Alignment with Sri Lanka's vision for smart and sustainable agriculture

Chapter 02- Business Case

2.1. Business Need

The current manual fruit inspection process in rural Sri Lankan agriculture creates multiple inefficiencies and cost implications:

- Manual labor costs: Fruit quality assessment requires significant manual labor time per harvest, diverting resources from other farming activities
- Quality inconsistency: Human error in ripeness assessment leads to premature or delayed harvesting directly impacting fruit quality and market value
- Disease spread: Delayed disease detection allows diseases to spread rapidly within crops increasing yield losses
- Accessibility gap: Existing technological solutions require high computational resources, cloud connectivity and technical expertise that are unavailable to most small-scale farmers
- Economic impact: Loss due to improper harvesting and undetected diseases directly affects farmer income and agricultural sector productivity

2.2. Business Objectives

This project establishes measurable objectives aligned with agricultural modernization goals:

- Develop an offline capable mobile application enabling farmers to assess fruit ripeness and disease status in real-time without internet dependency
- Achieve ≥95% accuracy on pomegranate detection and ripeness classification for Sri Lankan varieties under field conditions
- Optimize machine learning models to run efficiently on mid-range Android devices with latency <500ms per inference
- Create a reusable and localized dataset of 1000+ annotated pomegranate images representing multiple ripeness stages and disease conditions
- Deliver an affordable solution zero direct cost to farmers that demonstrably reduces fruit assessment time and improves decision making quality
- Establish a foundation for future agricultural AI applications in Sri Lanka through open-source dataset and model contributions

Chapter 03- Project Objectives

This project focuses on delivering specific technical and practical outcomes:

3.1. Functional Deliverables

- Develop and optimize lightweight machine learning models (YOLOv8, MobileNet/EfficientNet) for detecting pomegranate fruits, classifying ripeness stages (unripe, semi-ripe, ripe), and identifying disease symptoms
- Integrate trained models into a Flutter based mobile application with on-device inference using TensorFlow Lite.
- Implement augmented reality (AR) visualization interface overlaying ripeness and disease information directly on the live camera feed
- Create a custom dataset of 500–1000 annotated images of Sri Lankan pomegranate varieties captured under varied lighting, weather and field conditions

3.2. Quality Criteria and Success Metrics

- **Detection accuracy:** Grater than 95% precision and recall for pomegranate fruit detection on validation dataset
- **Classification accuracy:** Grater than 90% accuracy for ripeness stage classification
- **Inference performance:** Real-time performance (≥20 FPS) on mid-range Android devices
- **Latency:** Average inference latency <500ms per frame on target devices

- **Offline functionality:** Complete operation without cloud connectivity or internet requirements
- **User experience:** Intuitive AR interface requiring minimal training for field users

3.3. Acceptance Criteria

- Application successfully runs on Android 10+ devices with 4GB+ RAM
- Model files optimized to <100MB combined size for practical on-device deployment
- AR overlays display accurately aligned with detected fruits and provide actionable information
- Custom dataset includes diverse lighting conditions, growth stages, and disease variations
- Acceptable performance benchmarks and studies documentation

Chapter 04- Literature Review

4.1. Introduction and Search Strategy

This project builds upon significant recent advances in lightweight object detection, mobile machine learning, and agricultural automation. A systematic review of current literature was conducted using academic databases (Scopus, IEEE Xplore) and repositories (arXiv, Kaggle), focusing on keywords including "pomegranate detection," "lightweight YOLO," "mobile machine learning," "agricultural computer vision," and "TensorFlow Lite deployment."

4.2. Current Publications

Recent publications demonstrate the viability of lightweight deep learning architectures for agricultural applications:

- **Pomegranate-Specific Research:** Zhao et al. (2024) introduced YOLO-Granada, a lightweight YOLOv5/ShuffleNet model optimized for Android deployment that classifies pomegranate growth stages and detects disease symptoms in real-time. Xu et al. (2025) developed YOLO-MSNet, a pruned YOLOv11n model achieving notable accuracy improvements while maintaining reduced model size for efficient mobile deployment. Al Ansari (2024) demonstrated practical applicability by developing an AlexNet-based Android application for pomegranate leaf disease identification directly from mobile captured images.
- **Lightweight Model Architectures:** MobileNet and EfficientNet series have become standard baselines for mobile-constrained environments, offering favorable accuracy-efficiency trade-offs. YOLOv8 and recent YOLO variants provide state-of-the-art object detection with acceptable computational footprints for mobile deployment.

- **Mobile Machine Learning:** TensorFlow Lite has emerged as the standard framework for on-device inference, enabling low-latency, offline-capable applications. Quantization and pruning techniques can reduce model sizes by 75–90% with minimal accuracy loss.

4.3. Critical Analysis and Identified Gaps

While these studies confirm the technical feasibility of mobile-based pomegranate detection significant gaps remain:

1. **Regional Adaptation Gap:** Existing studies predominantly use datasets from other regions and do not account for the unique visual characteristics of Sri Lankan pomegranate varieties (Lanka Red, Malee Pink). These varieties differ substantially in color, texture, and shape from globally available datasets, limiting model generalization to local contexts.
2. **Dataset Localization Gap:** Most research relies on pre-trained models or synthetic augmentation without developing comprehensive region-specific datasets. This limits performance and reliability when deployed in Sri Lankan agricultural environments with distinct lighting conditions, soil types, and pest profiles.
3. **User Experience Gap:** Current systems emphasize detection accuracy but provide minimal consideration of practical farmer usability. Few implementations integrate intuitive AR interfaces or user-centered design principles for non-technical agricultural users.
4. **Deployment Validation Gap:** Limited real-world field testing and performance benchmarking exist for low-resource environments typical of Sri Lankan farming communities. Most evaluations occur in controlled laboratory settings.
5. **Methodological Gap:** Existing models depend on image augmentation and general-purpose datasets rather than diverse natural field captures representing genuine variability in ripeness stages and disease manifestations.

4.4. Implications for Project Design

This project directly addresses identified gaps through:

- Development of a localized dataset featuring Sri Lankan pomegranate varieties under authentic field conditions
- Implementation of region-specific model fine-tuning using transfer learning from pre-trained architectures
- Integration of user-friendly AR visualization for practical farmer accessibility
- Field based performance evaluation and validation in real agricultural environments
- Documentation of practical deployment considerations and latency benchmarks on target mid-range devices

Chapter 05- Method of Approach

5.1. Research Design

This project employs a prototyping and experimental research design, combining dataset creation, model development, and application implementation with iterative validation and performance evaluation.

5.2. Data Collection and Preparation

Dataset Scope:

- Primary: 500–800 images of Sri Lankan pomegranate varieties (Lanka Red, Malee Pink) captured under diverse field conditions
- Secondary: Complementary images from public Kaggle datasets for augmentation and generalization
- Total target: 2000+ annotated images post-augmentation

Collection Methodology:

- Image captured under varied lighting conditions (morning, midday, evening sunlight)
- Multiple weather conditions (clear, cloudy etc.)
- Diverse backgrounds (sky, trees, ground) representing authentic field scenarios
- Manual labeling of ripeness categories (unripe, semi-ripe, ripe) and disease type
- Compliance with image consent and data privacy protocols in collaboration with the Department of Agriculture

Preprocessing Pipeline:

- Fruit detection to isolate pomegranate regions
- Image normalization and alignment for consistent model input
- Data augmentation techniques: rotation, scaling, brightness adjustments, horizontal flipping
- Train/validation/test split: 70% / 20% / 10%

5.3. Algorithms, Tools, and Frameworks

Object Detection:

- YOLOv8 for real-time fruit localization and bounding box generation

Classification:

- MobileNetV2 or EfficientNet for ripeness classification and disease identification
- Transfer learning with fine-tuning on custom Sri Lankan dataset

Model Optimization:

- TensorFlow Lite (TFLite) conversion to reduce model size and inference latency
- TensorFlow Model Optimization Toolkit for pruning and quantization
- ONNX export for framework flexibility

Mobile Development:

- Flutter framework for cross-platform mobile application
- TensorFlow Lite Flutter plugin for on-device inference
- ARCore for augmented reality visualization

Development Stack:

- Python with TensorFlow, PyTorch, OpenCV
- Jupyter Notebook for model experimentation and analysis
- VS Code with Dart/Flutter extensions for mobile development
- Git for version control and GitHub for repository management

5.4. Evaluation Metrics and Validation Strategy

Model Evaluation:

- Precision, Recall, F1-Score for fruit detection
- Accuracy, Confusion Matrix for ripeness classification

Field Validation:

- Real world performance testing on mid-range Android devices
- Frame rate (FPS) measurement during live camera inference
- User acceptance testing with sample farmers or agricultural extension officers

Performance Benchmarking:

- Model size analysis
- Memory consumption on target devices
- Battery impact assessment
- Robustness testing under various environmental conditions

5.5. Project Management Methodology

This project will adopt Agile development practices:

- Phase 1 (Nov–Dec 2025): Dataset creation, annotation, and augmentation
- Phase 2 (Jan–Feb 2026): Model development, training, and fine-tuning
- Phase 3 (Mar–Apr 2026): Model optimization (TFLite conversion)
- Phase 4 (May–Jun 2026): Flutter application development and integration
- Phase 5 (Jul–Aug 2026): Field testing, validation, and refinement
- Phase 6 (Sep–Oct 2026): Documentation, benchmarking, and project completion

Chapter 06- Conceptual Diagram

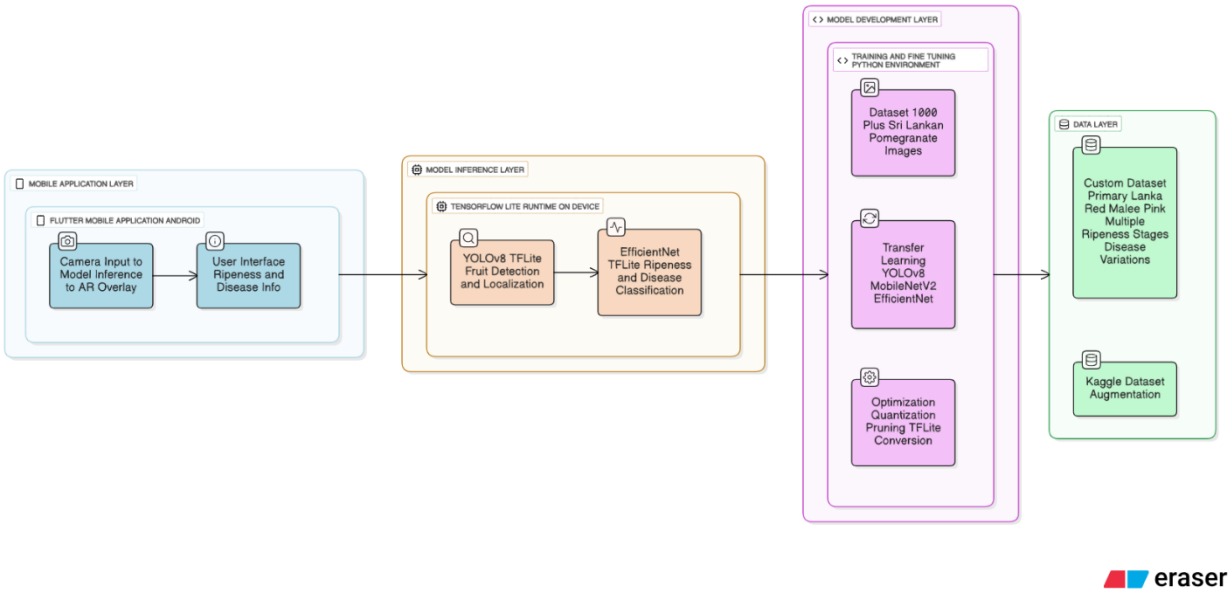

Figure: High level Architecture Diagram.

Chapter 07- Initial Project Plan

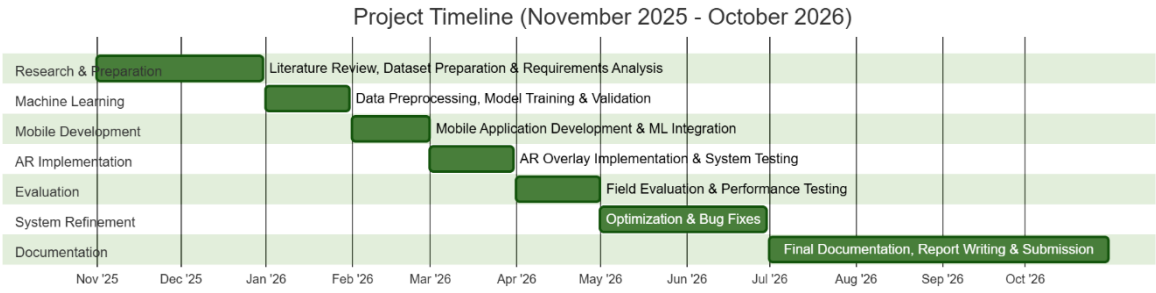

Figure: Project Gantt chart illustrating the timeline for pomegranate ripeness and disease detection system.

Chapter 08 Risk Analysis

Risk 1: Dataset Quality and Quantity Challenges

Description: Insufficient collection of diverse pomegranate images or poor annotation quality may limit model generalization to real-world field conditions.
Likelihood: Medium | Impact: High
Mitigation Strategies:

- Establish clear annotation guidelines and conduct inter-rater reliability checks
- Collaborate with Department of Agriculture to access diverse cultivar samples
- Implement data augmentation techniques to artificially increase dataset diversity
- Set minimum quality standards (image resolution ≤ 720p, clear fruit visibility)
- Early pilot testing with 100 images to validate annotation process before scaling

Risk 2: Model Accuracy Under Field Conditions

Description: Trained models may not generalize well to uncontrolled field lighting, occlusion and environmental variability despite training on augmented data.
Likelihood: Medium | Impact: High
Mitigation Strategies:

- Incorporate images captured under diverse natural lighting conditions during dataset collection
- Implement ensemble learning combining YOLOv8 and alternative detection architectures
- Conduct extensive field testing with real agricultural scenarios
- Establish fallback mechanisms (e.g., user confirmation for borderline classifications)
- Plan for model retraining with field feedback data if accuracy drops below thresholds

Risk 3: Mobile Device Performance Constraints

Description: Optimized models may exceed latency/memory targets on mid-range Android devices, compromising real-time performance.
Likelihood: Medium | Impact: High
Mitigation Strategies:

- Early latency profiling on target hardware during development phases
- Aggressive model quantization and pruning strategies
- Implementation of frame skipping or reduced resolution processing if needed
- Consider dual-model approach (fast detection + slower classification)
- Maintain performance benchmarking dashboard throughout development

Risk 4: Hardware and Infrastructure Failures

Description: Loss of training data, corruption of trained models or development environment crashes may disrupt progress.
Likelihood: Low | Impact: Medium
Mitigation Strategies:

- Implement daily automated backups to cloud storage (Google Drive, GitHub)
- Maintain version control with regular commits and tagged releases
- Document all training configurations and hyperparameters for reproducibility
- Use cloud-based Jupyter notebooks (Google Colab) with GPU access as backup training environment
- Maintain redundant copies of datasets across multiple storage locations

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